OR4023 (Optimization) Homework 4

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Question 1. Let

$$f(x_1, x_2) = 2x_1^2 + x_2^2 - 2x_1x_2 + 2x_1^3 + x_1^4.$$

Suppose that the function is minimized starting from $\mathbf{x}_0 = (0, -2)^T$. Verify that $\mathbf{p}_0 = (0, 1)^T$ is a direction of descent.

Solution. First, let's compute the gradient of f:

$$\nabla f(x_1, x_2) = \begin{pmatrix} 4x_1 + 6x_1^2 + 4x_1^3 - 2x_2 \\ 2x_2 - 2x_1 \end{pmatrix}$$

At $\mathbf{x}_0 = (0, -2)^T$, the gradient is:

$$\nabla f(\mathbf{x}_0) = \begin{pmatrix} 4 \\ -4 \end{pmatrix}$$

Now,

$$\nabla f(\mathbf{x}_0)^T \mathbf{p}_0 = \begin{pmatrix} 4 & -4 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = -4 < 0$$

Hence $\mathbf{p}_0 = (0, 1)^T$ is indeed a direction of descent at $\mathbf{x}_0$.

Question 2. Consider the quadratic function

$$f(\mathbf{x}) = \frac{1}{2}\mathbf{x}^T\mathbf{A}\mathbf{x} - \mathbf{c}^T\mathbf{x},$$

where $\mathbf{A}$ is a positive-definite matrix. Let $\mathbf{p}$ be a direction of descent for f at the point $\mathbf{x}$. Prove that the solution of the exact line search problem

$$\min_{\alpha > 0} f(\mathbf{x} + \alpha\mathbf{p})$$

is

$$\alpha = -\frac{\nabla f(\mathbf{x})^T\mathbf{p}}{\mathbf{p}^T\mathbf{A}\mathbf{p}}.$$

Solution. To find the optimal step size α, we need to minimize $f(\mathbf{x} + \alpha\mathbf{p})$ with respect to α. Let's solve this step by step:

1) First, let's expand $f(\mathbf{x} + \alpha\mathbf{p})$ using the quadratic form:

$$\phi(\alpha) = f(\mathbf{x} + \alpha\mathbf{p}) = \frac{1}{2}(\mathbf{x} + \alpha\mathbf{p})^T\mathbf{A}(\mathbf{x} + \alpha\mathbf{p}) - \mathbf{c}^T(\mathbf{x} + \alpha\mathbf{p})$$

2) Expanding the quadratic term:

$$= \frac{1}{2}(\mathbf{x}^T\mathbf{A}\mathbf{x} + \alpha\mathbf{x}^T\mathbf{A}\mathbf{p} + \alpha\mathbf{p}^T\mathbf{A}\mathbf{x} + \alpha^2\mathbf{p}^T\mathbf{A}\mathbf{p}) - \mathbf{c}^T\mathbf{x} - \alpha\mathbf{c}^T\mathbf{p}$$

3) Note that $\mathbf{x}^T\mathbf{A}\mathbf{p} = \mathbf{p}^T\mathbf{A}\mathbf{x}$ since $\mathbf{A}$ is symmetric:

$$= \frac{1}{2}\mathbf{x}^T\mathbf{A}\mathbf{x} + \alpha\mathbf{p}^T\mathbf{A}\mathbf{x} + \frac{1}{2}\alpha^2\mathbf{p}^T\mathbf{A}\mathbf{p} - \mathbf{c}^T\mathbf{x} - \alpha\mathbf{c}^T\mathbf{p}$$

$$= \frac{1}{2}\alpha^2 \mathbf{p}^T \mathbf{A}\mathbf{p} + \alpha(\mathbf{p}^T \mathbf{A}\mathbf{x} - \mathbf{c}^T \mathbf{p}) + \frac{1}{2}\mathbf{x}^T \mathbf{A}\mathbf{x} - \mathbf{c}^T \mathbf{x}$$

4) To find the minimum, we take the derivative with respect to α and set it to zero:

$$\phi'(\alpha) = \alpha \mathbf{p}^T \mathbf{A}\mathbf{p} + \mathbf{p}^T \mathbf{A}\mathbf{x} - \mathbf{c}^T \mathbf{p} = 0$$

5) Recall that $\nabla f(\mathbf{x}) = \mathbf{A}\mathbf{x} - \mathbf{c}$, therefore:

$$\mathbf{p}^T \nabla f(\mathbf{x}) + \alpha \mathbf{p}^T \mathbf{A}\mathbf{p} = 0$$

6) Solving for α:

$$\alpha = -\frac{\mathbf{p}^T \nabla f(\mathbf{x})}{\mathbf{p}^T \mathbf{A}\mathbf{p}} = -\frac{\nabla f(\mathbf{x})^T \mathbf{p}}{\mathbf{p}^T \mathbf{A}\mathbf{p}}$$

7) This is indeed a minimum (not a maximum) because the second derivative is $\mathbf{p}^T \mathbf{A}\mathbf{p} > 0$ (since $\mathbf{A}$ is positive definite).

Therefore, we have proven that $\alpha = -\frac{\nabla f(\mathbf{x})^T \mathbf{p}}{\mathbf{p}^T \mathbf{A}\mathbf{p}}$ is the optimal step size.

Question 3. Consider the problem

$$\text{minimize } f(x_1, x_2) = x_1^2 + 2x_2^2.$$

(a) If the starting point is $\mathbf{x}_0 = (2, 1)^T$, show that the sequence of points generated by the steepest-descent algorithm is given by

$$\mathbf{x}_k = \left(\frac{1}{3}\right)^k \begin{pmatrix} 2 \\ (-1)^k \end{pmatrix}$$

if an exact line search is used.

(b) Show that $f(\mathbf{x}_{k+1}) = f(\mathbf{x}_k)/9$.

(c) Compare the results in (b) to the bounds on the convergence rate of the steepest descent method when minimizing a quadratic function. What conclusions can you draw regarding this method?

Solution. (a) We want to find the steepest-descent updates for

$$f(x_1, x_2) = x_1^2 + 2x_2^2$$

starting from $\mathbf{x}_0 = (2, 1)^T$.

The gradient is

$$\nabla f(x_1, x_2) = (2x_1, 4x_2)^T.$$

An exact line search for a quadratic function yields

$$\alpha_k = \frac{\|\nabla f(\mathbf{x}_k)\|^2}{\nabla f(\mathbf{x}_k)^T A \nabla f(\mathbf{x}_k)},$$

where A is the Hessian of f.

For this problem:

$$A = \begin{pmatrix} 2 & 0 \\ 0 & 4 \end{pmatrix}.$$

Consequently, if $\mathbf{x}_k = (x_{1,k}, x_{2,k})^T$, then:

$$\nabla f(\mathbf{x}_k) = (2x_{1,k}, 4x_{2,k})^T,$$

$$\|\nabla f(\mathbf{x}_k)\|^2 = 4x_{1,k}^2 + 16x_{2,k}^2,$$

$$\nabla f(\mathbf{x}_k)^T A \nabla f(\mathbf{x}_k) = (2x_{1,k}, 4x_{2,k}) \begin{pmatrix} 2 & 0 \\ 0 & 4 \end{pmatrix} \begin{pmatrix} 2x_{1,k} \\ 4x_{2,k} \end{pmatrix} = 8x_{1,k}^2 + 64x_{2,k}^2.$$

Thus:

$$\alpha_k = \frac{4x_{1,k}^2 + 16x_{2,k}^2}{8x_{1,k}^2 + 64x_{2,k}^2}.$$

Starting with $\mathbf{x}_0 = (2,1)^T$:

$$\nabla f(\mathbf{x}_0) = (4,4)^T, \quad \alpha_0 = \frac{1}{3}.$$

Hence,

$$\mathbf{x}_1 = \mathbf{x}_0 - \alpha_0 \nabla f(\mathbf{x}_0) = (2,1)^T - \tfrac{1}{3}(4,4)^T = \left(\tfrac{2}{3}, -\tfrac{1}{3}\right)^T.$$

Observe that

$$\mathbf{x}_1 = \left(\tfrac{1}{3}\right)^1 \begin{pmatrix} 2 \\ -1 \end{pmatrix}.$$

Let us prove by induction that for every $k \geq 0$:

$$\mathbf{x}_k = \left(\tfrac{1}{3}\right)^k \begin{pmatrix} 2 \\ (-1)^k \end{pmatrix}.$$

The base case $k = 0$ holds since $\mathbf{x}_0 = (2,1)^T$. For $k = 1$, we already verified above that $\mathbf{x}_1 = (\frac{2}{3}, -\frac{1}{3})^T$.

Now assume the formula holds for some $k \geq 1$. Then:

$$\nabla f(\mathbf{x}_k) = \left(2 \cdot 2\left(\tfrac{1}{3}\right)^k, 4 \cdot (-1)^k \left(\tfrac{1}{3}\right)^k\right)^T = 4\left(\tfrac{1}{3}\right)^k \begin{pmatrix} 1 \\ 1(-1)^k \end{pmatrix}.$$

Therefore:

$$\|\nabla f(\mathbf{x}_k)\|^2 = 32\left(\tfrac{1}{3}\right)^{2k},$$

$$\nabla f(\mathbf{x}_k)^T A \nabla f(\mathbf{x}_k) = (\frac{1}{3})^{2k}(32 + 64) = 96 \left(\frac{1}{3}\right)^{2k}.$$

Thus:

$$\alpha_k = \frac{32}{96} = \frac{1}{3},$$

and:

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \alpha_k \nabla f(\mathbf{x}_k) = \left(\tfrac{1}{3}\right)^k \begin{pmatrix} 2 \\ (-1)^k \end{pmatrix} - \tfrac{1}{3} \cdot 4\left(\tfrac{1}{3}\right)^k \begin{pmatrix} 1 \\ 1(-1)^k \end{pmatrix} = \left(\tfrac{1}{3}\right)^{k+1} \begin{pmatrix} 2 \\ (-1)^{k+1} \end{pmatrix},$$

completing the induction.

(b) We calculate

$$f(\mathbf{x}_k) = x_{1,k}^2 + 2\,x_{2,k}^2 = 6\Big(\tfrac{1}{9}\Big)^k, \quad f(\mathbf{x}_{k+1}) = 6\Big(\tfrac{1}{9}\Big)^{k+1} = \frac{f(\mathbf{x}_k)}{9}.$$

(c) From standard theory for the steepest descent method on a quadratic, the convergence is linear with rate not greater than

$$\left(\frac{\lambda_{\max} - \lambda_{\min}}{\lambda_{\max} + \lambda_{\min}}\right)^2,$$

where $\lambda_{\min}$ and $\lambda_{\max}$ are the smallest and largest eigenvalues of the Hessian matrix. Here, $\lambda_{\min} = 2$ and $\lambda_{\max} = 4$, so:

$$\left(\frac{4-2}{4+2}\right)^2 = \left(\frac{1}{3}\right)^2 = \frac{1}{9}.$$

We found that

$$\frac{f(\mathbf{x}_{k+1})}{f(\mathbf{x}_k)} = \frac{1}{9},$$

which exactly matches this bound. Therefore, for this particular problem, the steepest descent method converges at precisely the worst-case rate predicted by theory.

Question 4. Apply Newton's method to solve $f(x) = x^2 - a = 0$, where $a > 0$. This is a good way to compute $\pm\sqrt{a}$. How does the iteration behave if $a \leq 0$?

Solution. To solve the equation

$$f(x) = x^2 - a = 0$$

using Newton's method, recall that the Newton update is

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}.$$

Therefore,

$$x_{k+1} = x_k - \frac{x_k^2 - a}{2x_k} = \frac{2x_k^2 - (x_k^2 - a)}{2x_k} = \frac{x_k^2 + a}{2x_k} = \frac{1}{2}\left(x_k + \frac{a}{x_k}\right).$$

Let's examine three cases with specific examples:
Case 1: When $a = 2$ (positive), starting with $x_0 = 1$:

$$\begin{aligned}
x_1 &= \frac{1}{2}\left(1 + \frac{2}{1}\right) = \frac{3}{2} = 1.5000 \\
x_2 &= \frac{1}{2}\left(1.5000 + \frac{2}{1.5000}\right) = \frac{2.2500 + 1.3333}{2} = 1.4167 \\
x_3 &= \frac{1}{2}\left(1.4167 + \frac{2}{1.4167}\right) = 1.4142 \\
x_4 &= \frac{1}{2}\left(1.4142 + \frac{2}{1.4142}\right) = 1.4142
\end{aligned}$$

The sequence converges rapidly to $\sqrt{2} \approx 1.4142$

Case 2: When $a = 0$, starting with $x_0 = 1$:

$$x_1 = \frac{1}{2}\left(1 + \frac{0}{1}\right) = 0.5000$$
$$x_2 = \frac{1}{2}\left(0.5000 + \frac{0}{0.5000}\right) = 0.2500$$
$$x_3 = \frac{1}{2}\left(0.2500 + \frac{0}{0.2500}\right) = 0.1250$$
$$x_4 = \frac{1}{2}\left(0.1250 + \frac{0}{0.1250}\right) = 0.0625$$

Each iteration halves the previous value, converging to 0.

Case 3: When $a = -2$ (negative), starting with $x_0 = -1$:

$$x_1 = \frac{1}{2}\left(-1 + \frac{-2}{-1}\right) = \frac{-1+2}{2} = 0.5000$$
$$x_2 = \frac{1}{2}\left(0.5000 + \frac{-2}{0.5000}\right) = \frac{0.5000 - 4.0000}{2} = -1.7500$$
$$x_3 = \frac{1}{2}\left(-1.7500 + \frac{-2}{-1.7500}\right) = \frac{-1.7500 + 1.1429}{2} = -0.3036$$
$$x_4 = \frac{1}{2}\left(-0.3036 + \frac{-2}{-0.3036}\right) = -3.2893$$

The sequence does not converge.

Question 5. Consider the problem

$$\min\ f(x_1, x_2) = (x_1 - 2x_2)^2 + x_1^4.$$

(a) Apply the steepest descent method with initial guess $(2, 1)^T$ for 2 iterations.

(b) Apply the classical Newton's method with initial guess $(2, 1)^T$ for 2 iterations.

Solution. (b) **Newton's Method** We first compute the gradient:

$$\nabla f(x_1, x_2) = \begin{pmatrix} 2(x_1 - 2x_2) + 4x_1^3 \\ -4(x_1 - 2x_2) \end{pmatrix}$$

we also need the Hessian matrix:

$$H(x_1, x_2) = \begin{pmatrix} 2 + 12x_1^2 & -4 \\ -4 & 8 \end{pmatrix}$$

At $\mathbf{x}_0 = (2, 1)^T$:

$$H(\mathbf{x}_0) = \begin{pmatrix} 2 + 12(2^2) & -4 \\ -4 & 8 \end{pmatrix} = \begin{pmatrix} 2 + 48 & -4 \\ -4 & 8 \end{pmatrix} = \begin{pmatrix} 50 & -4 \\ -4 & 8 \end{pmatrix}$$

We see the Hessian is positive definite, so the method is well-defined at this step. The Newton update is:

$$\mathbf{x}_{k+1} = \mathbf{x}_k - [H(\mathbf{x}_k)]^{-1} \nabla f(\mathbf{x}_k)$$

Computing the inverse of $H(\mathbf{x}_0)$:

$$[H(\mathbf{x}_0)]^{-1} = \frac{1}{50 \cdot 8 - (-4)^2} \begin{pmatrix} 8 & 4 \\ 4 & 50 \end{pmatrix} = \frac{1}{384} \begin{pmatrix} 8 & 4 \\ 4 & 50 \end{pmatrix}$$

Therefore:

$$\mathbf{x}_1 = \begin{pmatrix} 2 \\ 1 \end{pmatrix} - \frac{1}{384} \begin{pmatrix} 8 & 4 \\ 4 & 50 \end{pmatrix} \begin{pmatrix} 32 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix} - \begin{pmatrix} \frac{2}{3} \\ \frac{1}{3} \end{pmatrix} = \begin{pmatrix} \frac{4}{3} \\ \frac{2}{3} \end{pmatrix}$$

At $\mathbf{x}_1 = (\frac{4}{3}, \frac{2}{3})^T$:

$$\nabla f(\mathbf{x}_1) = \begin{pmatrix} 2(\frac{4}{3} - 2\frac{2}{3}) + 4(\frac{4}{3})^3 \\ -4(\frac{4}{3} - 2\frac{2}{3}) \end{pmatrix} = \begin{pmatrix} 9.4815 \\ 0 \end{pmatrix}$$

$$H(\mathbf{x}_1) = \begin{pmatrix} 2 + 12(\frac{4}{3})^2 & -4 \\ -4 & 8 \end{pmatrix} = \begin{pmatrix} 23.3333 & -4 \\ -4 & 8 \end{pmatrix}$$

The Hessian is positive definite, so the method is well-defined at this step. Computing the second Newton step:

$$\mathbf{x}_2 = \mathbf{x}_1 - [H(\mathbf{x}_1)]^{-1} \nabla f(\mathbf{x}_1) = \begin{pmatrix} 0.8889 \\ 0.4444 \end{pmatrix}$$